Denoising and SNR-Aware SpeechCueLLM for Robust Speech Emotion Recognition

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I. Project Details

Criterion followed: If code is available, reproduce the existing implementation and address an identified limitation through a proposed and experimentally evaluated solution.

The project is based on *SpeechCueLLM*, "Beyond Silent Letters: Amplifying LLMs in Emotion Recognition with Vocal Nuances," published in the Findings of ACL: NAACL 2025 [1]. SpeechCueLLM converts five acoustic characteristics (average volume, volume variation, average pitch, pitch variation, and speaking rate) into categorical natural-language descriptions using a deterministic quantile-based pipeline, and supplies these together with textual conversational context to an LLM. The original authors evaluate on IEMOCAP and MELD and provide an official implementation [2], which makes reproduction of the baseline feasible.

Paper: https://aclanthology.org/2025.findings-naacl.117/
Official implementation: https://github.com/zehuiwu/SpeechCueLLM

The original paper reports strong gains on the clean, studio-quality IEMOCAP dataset but only marginal (sometimes negative) gains on MELD, whose audio contains background noise, overlapping speakers, and short clips. The authors themselves attribute this to unreliable acoustic feature extraction under noisy conditions and list improved handling of noisy, real-world audio as future work. This project directly targets that gap with two concrete, connected contributions: (1) a **denoising and SNR-aware acoustic description pipeline** that cleans the input audio and explicitly communicates measurement reliability to the LLM, and (2) a **parameter-efficient fine-tuning (PEFT) comparison** (QLoRA, LoRA rank sweep, DoRA) layered on top of the improved pipeline to study the accuracy–efficiency trade-off.

Team Composition and Individual Contributions:

- **Member 1:** Utkarsh Umap, B24CS049, umapus@iitbhilai.ac.in. Reproduces the official SpeechCueLLM implementation, prepares datasets and environment, verifies preprocessing and baseline results, integrates the denoising front-end into the pipeline, and maintains the main experimental pipeline.
- **Member 2:** Kaif Raza, B24DS017, mdkraza@iitbhilai.ac.in. Implements blind SNR estimation, designs the SNR-aware confidence-modulated description generation stage, builds the extended prompt format, and conducts description-level ablations (B2 vs B3).
- **Member 3:** Kushal Agarwal, B24DS012, kushal@iitbhilai.ac.in. Implements and compares PEFT methods (LoRA rank sweep, QLoRA, DoRA), measures efficiency metrics (memory, training time, latency), and performs robustness evaluation across noise conditions and statistical analysis.

TABLE I
Approximate project workload distribution

Work Area	Weight
Baseline reproduction, dataset preparation and verification	30%
Denoising front-end and SNR-aware description pipeline	35%
PEFT efficiency comparison (QLoRA, rank sweep, DoRA)	25%
Joint analysis, documentation and presentation	10%
Total	**100%**

II. Problem Statement

Context and motivation: Speech Emotion Recognition (SER) identifies emotional states conveyed through spoken language and is relevant to conversational agents, human–computer interaction, customer-service systems, and educational technologies. A transcript alone does not preserve vocal cues such as pitch, loudness, or speaking rate that often carry emotional meaning independent of the words used. SpeechCueLLM addresses this by converting acoustic measurements into natural-language descriptions inserted directly into the LLM prompt, without modifying the LLM's architecture. However, this conversion pipeline silently assumes that the underlying acoustic measurements are reliable. When the input audio is noisy, that assumption breaks down: the numbers fed into the pipeline are already corrupted, and the pipeline still produces a confident-sounding textual description, misleading the LLM.

Research problem: How can SpeechCueLLM be made robust to noisy, real-world audio such as MELD by (a) reducing noise in the audio before feature extraction, and (b) explicitly telling the LLM how reliable each acoustic description is, so it can discount unreliable cues rather than trusting them at face value?

Research gap: SpeechCueLLM's four-stage audio-to-text pipeline (threshold calculation, speaker-specific thresholding, categorization, description generation) is entirely rule-based and deterministic. It has no mechanism to detect or communicate when the raw pitch/volume/rate measurements it is built on are themselves unreliable due to noise. This project closes that gap directly, rather than replacing the feature set or the LLM architecture, keeping the original paper's "no architecture change, plain-text interpretability" philosophy intact while fixing its stated main weakness.

Research questions:

- **RQ1:** Can the official SpeechCueLLM implementation and its reported baseline behaviour be reproduced reliably on the available hardware?
- **RQ2:** Does adding a speech-enhancement (denoising) front-end before acoustic feature extraction improve SpeechCueLLM's performance on noisy audio (MELD) and on artificially degraded IEMOCAP?
- **RQ3:** Can a blind SNR estimate per utterance be used to modulate the confidence/wording of the pitch, volume, and speaking-rate descriptions, and does exposing this SNR information to the LLM (as an explicit description and as a confidence modulator) improve emotion recognition under degraded audio?
- **RQ4:** Among parameter-efficient fine-tuning methods (standard LoRA, LoRA rank sweep, QLoRA, DoRA) applied on top of the denoising + SNR-aware pipeline, which gives the best accuracy-per-trainable-parameter and accuracy-per-GPU-memory trade-off?
- **RQ5:** Does the combined denoising + SNR-aware pipeline improve robustness to noise without materially increasing computational cost relative to the original SpeechCueLLM prompt-based approach?

III. Methodology

The project proceeds in six phases: (1) reproduction and verification of SpeechCueLLM, (2) denoising front-end integration, (3) blind SNR estimation, (4) SNR-aware description generation, (5) parameter-efficient fine-tuning comparison, and (6) ablation and robustness evaluation. Figure 1 shows the full proposed pipeline stage-by-stage, and Table II compares each stage against the original SpeechCueLLM implementation.

Phase 1 – Reproduction: obtain IEMOCAP/MELD through official channels; recreate the SpeechCueLLM software environment; use the authors' supplied acoustic features and code for the first baseline; reproduce feature processing, categorical-description conversion, and prompt construction; run the original training/inference pipeline; evaluate with the original metrics. The published results report a weighted F1 of 70.11 for text-only LLaMA-2-7B and 72.02–72.6 after adding speech descriptions [1]; these are treated as reference points, not guaranteed targets.

Phase 2 – Denoising front-end: a speech-enhancement model is inserted before the original acoustic feature extractor, so that pitch, volume, and speaking-rate statistics are computed on enhanced rather than raw audio. SpeechBrain MetricGAN+ [5] is fixed as the denoising model for this project, chosen for its low compute footprint and ease of integration on a 16–24 GB GPU; the Facebook/Meta Denoiser [6] is noted as a possible future comparison but is not part of the core pipeline. The same denoised audio is used consistently for feature extraction in all denoising-enabled configurations.

Because denoising can shift the scale of a feature distribution rather than simply removing noise-driven outliers, a diagnostic step precedes any decision about the original pipeline's Stage 1 (threshold calculation): raw and denoised feature distributions (average/variation of pitch and volume, speaking rate) are compared across a representative sample of utterances from each dataset. If the distributions are closely aligned, the original thresholds (computed on raw audio, as in the published pipeline) are reused directly on denoised features (*frozen-threshold variant*). If a systematic shift is observed, thresholds are instead recomputed from scratch on the denoised feature distributions (*recomputed-threshold variant*). Both variants are retained as a controlled sub-experiment rather than assumed in advance, and the outcome of this diagnostic is reported explicitly.

Phase 3 – Blind SNR estimation: since paired clean/noisy references are not available for MELD, a reference-free SNR estimate is computed per utterance using WADA-SNR [7] (with a voice-activity-detection-based noise-floor estimate as a cross-check). This SNR value is the reliability signal used in Phase 4. SNR thresholds separating "low," "moderate," and "high" reliability bands are not fixed a priori; they are set empirically from the observed SNR distribution of each dataset (e.g., using dataset-specific quantiles), consistent with how the original paper derives its own feature thresholds.

Phase 4 – SNR-aware description generation: the original four-stage pipeline (threshold calculation, speaker-specific thresholding, categorization, description generation) is retained in structure, and extended in two ways using the SNR estimate from Phase 3, as illustrated in Fig. 1:

1) **Explicit reporting:** the estimated SNR is stated directly in the prompt as an additional description, e.g. "audio quality: moderate (SNR $\approx 12\,\mathrm{dB}$)".
2) **Confidence modulation:** the SNR value controls the wording of the pitch/volume/speaking-rate descriptions themselves – under low SNR, feature descriptions are hedged (e.g. "pitch appears high, but the measurement is unreliable under the current audio quality") rather than stated flatly.

To keep the prompt structure fixed and consistent across all examples (important for stable LoRA fine-tuning), every acoustic feature is *always* included in the description regardless of SNR; low-confidence measurements are hedged more heavily rather than omitted outright. The SNR signal describes measurement reliability only; it never invents or overrides the emotion label.

Phase 5 – Parameter-efficient fine-tuning comparison: on top of the best-performing pipeline configuration from Phase 4, three PEFT strategies are compared against the original paper's standard LoRA setup:

- **QLoRA** [8]: LoRA adapters trained on top of a 4-bit quantized frozen backbone, to measure GPU memory savings at comparable accuracy.
- **LoRA rank sweep**: rank $r \in \{4, 8, 16, 32\}$ swept on the original LoRA setup to trace a trainable-parameter-vs-F1 trade-off curve.
- **DoRA** [9]: weight-decomposed LoRA, compared against standard LoRA at matched rank to test whether it improves accuracy at the same parameter budget.

Phase 6 – Ablation and robustness evaluation: Table III lists the planned configurations. Robustness across clean, mild, moderate, and severe noise conditions is measured using

$$\Delta F1 = F1_{\mathrm{clean}} - F1_{\mathrm{degraded}}, \tag{1}$$

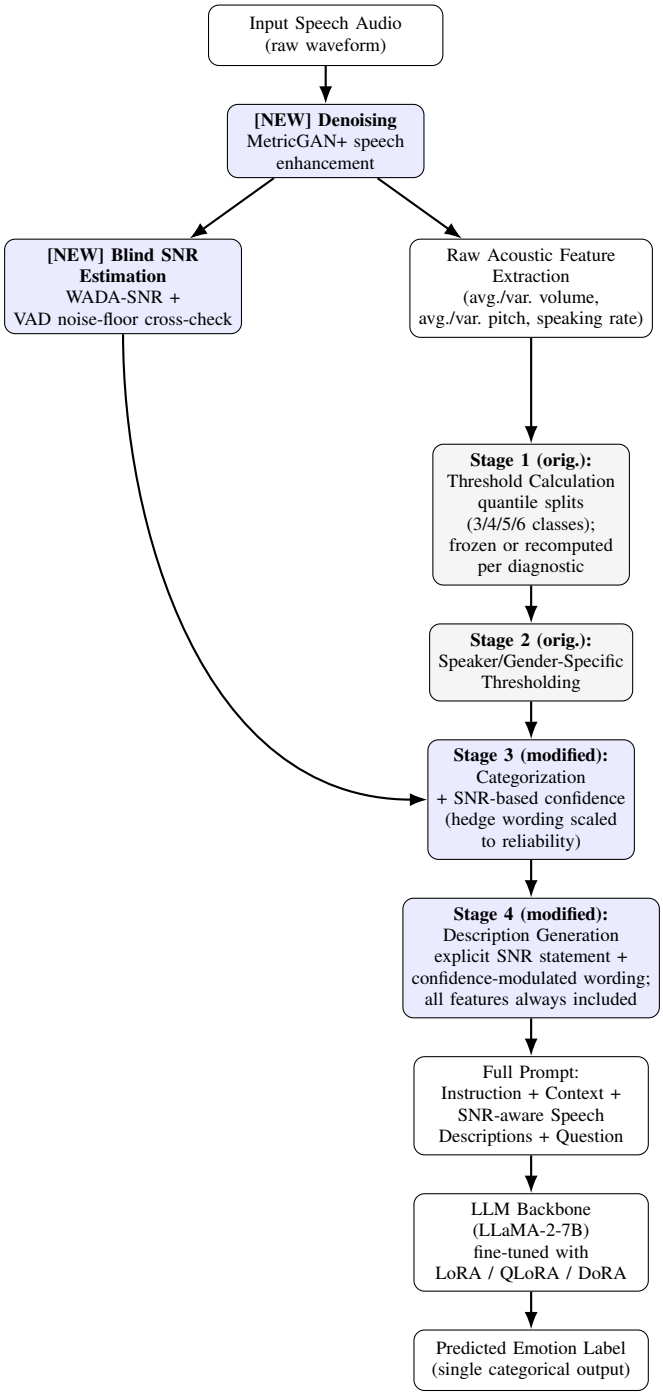


Fig. 1. Proposed pipeline. Gray boxes are unchanged from the original SpeechCueLLM four-stage pipeline (threshold calculation, speaker-specific thresholding, categorization, description generation); blue boxes are new or modified components. The SNR signal, estimated once per (denoised) utterance, is injected into Stages 3 and 4 without altering the internal logic of Stages 1–2.

$$\text{Relative Robustness} = \frac{F1_{\text{degraded}}}{F1_{\text{clean}}}. \tag{2}$$

The comparison between B1 and B2 isolates the effect of denoising alone; B2 vs. B3 isolates the effect of SNR-aware descriptions on top of denoising; B4/B5/B6 isolate the effect of each PEFT method on top of the best pipeline (B3).

Evaluation metrics: recognition – accuracy, macro-F1, weighted-F1, per-class F1 (macro-F1 emphasized due to MELD's class imbalance); robustness – macro/weighted-F1 across SNR levels, clean-to-noisy degradation, relative robustness; efficiency – trainable parameters, GPU memory, training time, inference latency, denoising and SNR-estimation preprocessing overhead. Where feasible, experiments are repeated across seeds and mean/standard deviation reported.

IV. Resource and Dataset Availability

IEMOCAP [3] is one of the two datasets directly evaluated by SpeechCueLLM and contains relatively high-quality con-

TABLE II
STAGE-BY-STAGE COMPARISON BETWEEN THE ORIGINAL SPEECHCUELLM PIPELINE AND THE PROPOSED EXTENSION

Pipeline Stage	Original SpeechCueLLM	Proposed (this project)
Pre-processing	Raw audio used directly for feature extraction.	Raw audio first passed through a MetricGAN+ denoising front-end; enhanced audio used for feature extraction.
Reliability signal	None – audio quality is not measured or communicated.	Blind per-utterance SNR estimated via WADA-SNR (cross-checked with a VAD-based noise-floor estimate).
Stage 1: Threshold Calculation	Global quantile-based thresholds per feature, computed once over the (raw-audio) dataset.	Thresholds frozen from raw-audio computation or recomputed on denoised features, decided per a raw-vs-denoised distribution diagnostic (Phase 2).
Stage 2: Speaker-Specific Thresholding	Falls back silently to global thresholds when speaker data is limited.	Unchanged mechanism; effect of denoising on threshold stability evaluated separately (B1 vs. B2).
Stage 3: Categorization	Hard bucket boundaries (e.g., very low/low/medium/high/very high); no notion of confidence.	Same bucket boundaries, but each categorization is additionally tagged with a confidence level derived from the SNR estimate.
Stage 4: Description Generation	Fixed phrase templates stated with uniform confidence (e.g., "high volume with moderate variation").	Phrase templates conditioned on SNR: explicit SNR statement added to the prompt, and hedge words scaled to measurement confidence; all features always included to preserve a fixed prompt structure.
Fine-tuning	LoRA only, fixed rank, on LLaMA-2/3, Phi-3.	LoRA (rank sweep), QLoRA, and DoRA compared on top of the improved pipeline.
Target weakness	Degrades sharply on noisy audio (MELD); authors list this as future work.	Directly targets this weakness via denoising + reliability-aware description generation.

TABLE III
PLANNED COMPARISON AND ABLATION CONFIGURATIONS

ID	Configuration
B0	Text-only LLM baseline
B1	Original SpeechCueLLM (reproduction)
B2	SpeechCueLLM + denoising front-end (no SNR in prompt)
B3	SpeechCueLLM + denoising + SNR-aware descriptions (explicit + confidence-modulated)
B4	B3 fine-tuned with QLoRA
B5	B3 fine-tuned with DoRA
B6	B3 with LoRA rank sweep ($r = 4, 8, 16, 32$)

TABLE IV
LIMITATION ADDRESSED AND PLANNED INVESTIGATION

Limitation	Planned Investigation
Dependence on clean audio; acoustic features become unreliable under noise (weak MELD results)	Insert a denoising front-end before feature extraction, and estimate per-utterance SNR to modulate/annotate the confidence of each acoustic description.
Fine-tuning cost of LoRA at fixed rank	Compare QLoRA, a LoRA rank sweep, and DoRA on top of the improved pipeline to map the accuracy–memory–parameter trade-off.

versational recordings suitable for baseline reproduction and for controlled artificial-noise experiments (https://sail.usc.edu/iemocap/), obtained and used under its official terms.

MELD [4] is the second evaluation dataset (https://affective-meld.github.io/); its real-world, noisy audio conditions make it the primary target for the proposed denoising and SNR-aware pipeline, since the original SpeechCueLLM benefit on MELD is small or negative.

Preprocessing: verify dataset structure and labels; align audio, transcripts, and annotations; reproduce the original SpeechCueLLM preprocessing as closely as possible; use the officially supplied feature files for the first baseline; run the denoising front-end on all audio used in denoising-enabled configurations; compute the blind SNR estimate per utterance; run the raw-vs-denoised feature distribution diagnostic to fix the Stage 1 thresholding strategy; generate SNR-aware descriptions; and keep train/test splits speaker- or dialogue-disjoint per the original protocol. No new manually labelled dataset is required.

Official resources: the SpeechCueLLM repository [2] supplies feature files and training/inference code; the environment and dependency versions specified in the official repository will be used for baseline reproduction before any modernization.

Software: Python, PyTorch, Hugging Face Transformers, the PEFT library (LoRA, QLoRA via bitsandbytes, DoRA), SpeechBrain (MetricGAN+), the Facebook/Meta Denoiser package, an SNR-estimation implementation (WADA-SNR), NumPy, pandas, scikit-learn, CUDA, Git/GitHub.

Hardware: a GPU with approximately 16–24 GB VRAM is expected to be sufficient, to be provided as needed. QLoRA's 4-bit quantization is expected to reduce memory requirements further for the corresponding experiments. Cloud GPU resources will be used for selected experiments if local/institutional resources are insufficient.

V. LIMITATIONS FOUND

Table IV summarizes the limitation of the original SpeechCueLLM study addressed by this project.

Main limitation addressed: the dependence of SpeechCueLLM's acoustic descriptions on clean input audio. The hypothesis is that (a) denoising the audio before feature extraction reduces measurement error at the source, and (b) explicitly communicating remaining measurement uncertainty via SNR-aware descriptions lets the LLM rely less on untrustworthy cues – both to be evaluated experimentally against the original pipeline, not assumed.

Why this work is unique: the contribution is a denoising-plus-SNR-aware extension of the exact four-stage textual pipeline used by SpeechCueLLM, combined with a systematic

comparison of modern parameter-efficient fine-tuning methods (QLoRA, DoRA, LoRA rank sweep) on top of it. No new LLM architecture, audio encoder, or embedding-fusion mechanism is introduced; the interpretable, text-only prompting paradigm of the original paper is preserved throughout.

VI. Expected Outcomes

Planned deliverables: a reproducible SpeechCueLLM baseline; a documented preprocessing/evaluation pipeline for IEMOCAP and MELD; a working denoising front-end integrated into acoustic feature extraction; a raw-vs-denoised feature distribution diagnostic and resulting thresholding decision; a blind per-utterance SNR estimator; an SNR-aware description generation module (explicit reporting and confidence modulation); a comparison of QLoRA, LoRA rank sweep, and DoRA on top of the improved pipeline; robustness experiments across noise levels; the B0–B6 ablation study; clean vs. degraded performance comparisons; computational-resource measurements (parameters, memory, training time, latency); trained checkpoints and reproducible configurations; tables, plots, and a final report with an optional system demonstration.

No fixed numerical F1 target is assumed. Success is judged along **accuracy, robustness, and computational cost**: improved macro/weighted-F1 on MELD and on artificially degraded IEMOCAP relative to the original pipeline (B1), lower clean-to-noisy degradation, a clear accuracy-per-trainable-parameter/memory trade-off curve across PEFT methods, and limited preprocessing overhead relative to the original pipeline. If the denoising or SNR-aware mechanisms do not help, the negative result and its analysis will be reported rather than an unsupported positive claim.

References

[1] Z. Wu, Z. Gong, L. Ai, P. Shi, K. Donbekci, and J. Hirschberg, "Beyond Silent Letters: Amplifying LLMs in Emotion Recognition with Vocal Nuances," in *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 2202–2218, 2025.

[2] Z. Wu *et al.*, *SpeechCueLLM: Official Implementation*, GitHub repository, 2025. [Online]. Available: https://github.com/zehuiwu/SpeechCueLLM

[3] C. Busso *et al.*, "IEMOCAP: Interactive Emotional Dyadic Motion Capture Database," *Language Resources and Evaluation*, vol. 42, pp. 335–359, 2008.

[4] S. Poria *et al.*, "MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations," in *Proc. 57th Annual Meeting of the ACL*, 2019.

[5] S.-W. Fu, C. Yu, T.-A. Hsieh, P. Plantinga, M. Ravanelli, X. Lu, and Y. Tsao, "MetricGAN+: An Improved Version of MetricGAN for Speech Enhancement," in *Proc. Interspeech 2021*, 2021.

[6] A. Défossez, G. Synnaeve, and Y. Adi, "Real Time Speech Enhancement in the Waveform Domain," in *Proc. Interspeech 2020*, 2020.

[7] C. Kim and R. M. Stern, "Robust Signal-to-Noise Ratio Estimation Based on Waveform Amplitude Distribution Analysis," in *Proc. Interspeech 2008*, 2008.

[8] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient Finetuning of Quantized LLMs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[9] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, and M.-H. Chen, "DoRA: Weight-Decomposed Low-Rank Adaptation," in *Proc. International Conference on Machine Learning (ICML)*, 2024.

Appendix

AI-assisted tools such as ChatGPT and Claude were used during the project to research and understand background concepts, including speech enhancement/denoising models, blind SNR estimation methods, parameter-efficient fine-tuning techniques (LoRA, QLoRA, DoRA), and the internal mechanics of the SpeechCueLLM pipeline. These tools were also used to clarify unfamiliar terminology, compare candidate denoising and PEFT methods, and assist with organizing and refining the technical writing of this proposal.

Some prompts used during research:

- "Explain the SpeechCueLLM paper step by step in simple terms."
- "What is the final output of the SpeechCueLLM model, and how is it evaluated?"
- "What role does LoRA play in this paper, and how does it differ from full fine-tuning?"
- "What are practical, lightweight speech-enhancement/denoising models suitable for a 16–24GB GPU?"
- "What are standard blind/reference-free SNR estimation methods for real-world noisy audio?"
- "Compare QLoRA, LoRA rank variation, and DoRA as parameter-efficient fine-tuning strategies."
- "How can an SNR estimate be used to modulate confidence in categorical feature descriptions fed to an LLM?"
- "Should acoustic-feature thresholds be recomputed on denoised audio or kept from the original raw-audio computation?"

Other anticipated platforms include GitHub for version control, VS Code (or equivalent), institutional/cloud GPU resources, and Python/PyTorch-based ML environments.